Geometry by Construction: A Geometric Prior Bottleneck

Anonymous authors

September 3, 2026

1 Introduction

The information bottleneck (IB) principle [Tishby et al., 2000] seeks a representation Z of an input X that retains the information necessary to predict a label Y while discarding the rest. Because the mutual information terms are intractable for deep networks, the variational information bottleneck (VIB) [Alemi et al., 2017] bounds the compression term $I(X;Z)$ with a fixed marginal prior $\mathcal{N}(0,I)$; the conditional entropy bottleneck (CEB) [Fischer, 2020] replaces the marginal prior with a trainable label-conditional backward encoder $b(z|y)$ and thereby sharpens the bound from the full rate to the *residual* information $I(X;Z|Y)$, certified by the gap $\mathrm{ReX} = \mathbb{E}[\mathrm{KL}(q(z|x)\|b(z|y))] \geq I(X;Z|Y)$. Nonlinear IB [Kolchinsky et al., 2017] and the multivariate variational formulation of [Abdelaleem et al., 2025] complete the deep-variational lineage. The certificate is the central object of this paper, and its failure modes are our starting point.

One background failure mode and one problem. In the deterministic scenario $Y = f(X)$ — the regime of classification benchmarks — the conditional bottleneck degenerates in two independent ways, of which the first is settled background and the second is the problem this paper addresses. First, *the knob is dead*: as shown by Kolchinsky and Tracey [2019] (their Caveat 1), every $\gamma > 0$ of the CEB objective selects the same corner of the information plane, $I(Y;Z) = H(Y)$ with $I(X;Z|Y) = 0$, because the CEB weight is $\beta_{\mathrm{IB}} = 1 + \gamma > 1$ and the deterministic IB curve is linear. The same paper provides the fix — a squared compression term, which restores a unique interior optimum per β — so we do not list this channel as a problem to solve; in Section 2.2 we show, instead, that our anchored reference supplies the same strict convexity by a different mechanism (a quadratic displacement term built into the KL by construction), as a connection to the Caveats fix, and we report the β-sweep empirically. Second, and less recognized, *the certificate is blind to the relationships between labels*: with the backward encoder matched to the posterior, the certificate collapses to the residual $I(X;Z|Y)$ whatever the pairwise geometry of the class-conditional priors — class means merged, coincident, or placed in arbitrary directions all yield the same certificate value. On the family $T_\alpha = B_\alpha \cdot Y$ that interpolates between the collapsed code and the exact copy of the label, the residual vanishes, so the certificate reads zero across the whole retention axis — tightness of the certificate is compatible with retaining all, part, or none of the label. We prove this channel in Propositions 2.2 and 2.3; closing it — by declaring the inter-class geometry in the reference — is the core contribution of this paper.

The missing ingredient. Channel 2 exposes a structural gap: the *rate/certificate* part of the CEB objective contains no cross-class term — the certificate is a sum of per-class matching terms, each sample touching only its own class prior — so the certificate itself never constrains the geometry *among* the class-conditional priors $\{b(z|y = k)\}$: at a fixed posterior, its value is invariant to the pairwise geometry of the priors. The prediction term can influence this geometry only indirectly, through the posterior, and no term of the objective couples two distinct class priors directly. Two class means may sit arbitrarily close, coincide, or lie in arbitrary directions without changing the certificate's value. CEB therefore compresses *against a reference with no declared structure*: the certificate measures per-class proximity to whatever reference the optimizer finds convenient, and records nothing about the relationships between labels. What a representation should look like when it is properly compressed — class means separated, ordered, or laid out on a frame — is exactly what the certificate leaves unspecified. Our framework turns such a structure into a declared prior. This is the gap the present paper closes — the unspecified geometry of the reference; it does not claim that the certificate meters retained label information (the precise scope is delimited in Section 2.2 and the appendix).

A geometric prior bottleneck. We propose a framework in which the label-conditional prior — the *reference* of the compression term — is confined to a geometric family by construction. The general principle is simple: the prior $p(z|y)$ must encode, through its geometry alone, the structure of the label space, and it must do so in a way imposed by construction at every step, rather than left to the optimizer. Formally, the family is *qualified* by two checkable conditions — a bounded covariance and a mean map that encodes the label metric at a fixed, declared scale (a two-sided separation condition for discrete labels, a two-sided metric condition for continuous labels; Section 2.2). For discrete labels the structure to encode is *class equality* — classes are either identical or distinct, nothing else — and for continuous labels it is *order and numerical distance*. Concretely, the prior is computed at every forward pass by a stateless geometric operation on the prior encoder's output: the resulting prior family cannot merge two labels, cannot collapse to a point, and cannot rescale its geometry — these degenerations of the *reference* are excluded by geometry rather than by the optimizer's good behavior. (The prior variance is the fixed shared value τ^2, Section 2.3.1.) We emphasize the scope of these constraints: they bind the reference, not the deployed representation. The posterior remains free — in particular, it can still realize a full-retention label code with a zero certificate, as the trainable frame rotates to meet it — so the framework constrains the geometric *form* of the label code and does not claim that the certificate meters retained label information. The sample-level posterior, the single-KL objective, and the residual-certificate structure of CEB are kept unchanged, so the entire analysis of the certificate transfers pointwise. We call this family of objectives a *geometric prior bottleneck* (GPB). Two instantiations realize the principle.

For classification, the prior encoder maps the *all*-class one-hot matrix to the raw class-mean matrix at every step, the matrix is orthonormalized by a thin polar factorization (exactly equivariant under class relabeling), and the prior means become the scaled orthogonal frame $a\,q_{\theta,k}$: every pair of distinct classes sits at distance $\sqrt{2}\,a$, with a class-mean Gram matrix equal to $a^2 I_K$ *by construction*. We call this variant the *orthogonal-prior bottleneck* (OPB). The orthogonal frame is a design choice of this instance, motivated — but not dictated — by the neural-collapse phenomenon, in which terminal-phase classifiers concentrate class means onto a simplex ETF, a *centered* regular simplex with pairwise cosines $-1/(K-1)$ [Papyan et al., 2020]. We choose the orthogonal frame instead of the simplex ETF because it expresses exactly the class-equality structure (all pairs equidistant), is permutation-equivariant, and is simpler to compute, at the cost of requiring $K \le d$; we do not claim to reproduce the neural-collapse geometry. For regression, the prior is a line: the prior means are $\rho\,\tilde{y}\,u$ for a unit direction u normalized from a trainable vector at every step, so that latent distances are exactly ρ times standardized label distances. We call this variant the *equidistant prior bottleneck* (EPB). Both variants are stateless — no EMA, no prototypes, no per-batch statistics — and both keep a single KL term.

The framework earns its keep analytically. At the *framework* level, two geometry-independent statements hold. *(i) A structured conditional prior*: any stateless geometric prior keeps the CEB certificate valid — the residual bound $I(X;Z|Y)$ remains in force at every parameter value, with a caveat that a batch-dependent orthonormalization would break it (Proposition 2.4). *(ii) Anchor alignment and separation transfer*: in the deterministic K-class setting, minimizing the GPB objective forces the expected anchor mismatch below $C_{\mathrm{G}}/\beta \le (\ln K)/\beta$ at any near-optimum, with $C_{\mathrm{G}} = H(Y|Z_{\mathrm{align}})$ the Bayes cross-entropy on the aligned mixture — attained by the main model's anchor-energy classifier in the orthogonal instance, where the covariance-determinant factors cancel (Proposition 2.6), and the posterior class centers then inherit the prior's separation — $\|m_i - m_j\| \ge \|m_\theta(i) - m_\theta(j)\|$ minus mismatch terms of order $1/\sqrt{\beta\pi}$ (Proposition 2.5; Corollary 2.1). In the deterministic regression setting the same comparison argument gives $\mathbb{E}[D(Y)] \le (\tau^2/\rho^2 + \varepsilon)/\beta$ (Proposition 2.7), so the pointwise transfer holds on label sets of measure at least $1 - (\tau^2/\rho^2 + \varepsilon)/(\beta t)$ (Corollary 2.2). At the *instance* level, the orthogonal frame supplies a constructive class geometry — any class-independent posterior mean pays a *positive compression cost* of at least $a^2(1-1/K)/(2\tau^2)$ for uniform classes, so the VIB corner $\mu \equiv 0$ is a penalized configuration rather than a free optimum (Proposition 2.9) — and the conditional posterior center separation guarantee instantiates the framework transfer with $\Delta = \sqrt{2}\,a$ (Corollary 2.3). Further results — the KL decomposition, a Fano-type label-information lower bound, and the strict convexity of the aligned stochastic surrogate — are collected in the appendix.

We evaluate the framework empirically across nine datasets and four backbones, and we verify the mechanism itself rather than only end-to-end accuracy: we measure the geometry of the learned priors and of the posterior class centers, test whether a parameter-free nearest-anchor rule can replace the trained classifier, and compare the main model's anchor-energy classifier — whose weights are the frame itself — against the free linear classifier. The experiments show that the orthogonal geometry transfers from the prior to the posterior, that the anchor scale directly controls the separation and cross-seed consistency of the representation, and that prediction quality does not require the free classifier's degrees of freedom.

Our contributions are: (1) the GPB framework, in which the label-conditional prior — the reference of the compression term — encodes the structure of the label space by construction (constraining the geometric form of the label code, not the retention of label information); (2) the OPB and EPB variants — a stateless orthogonal frame for classification and a strictly isometric axis for regression — with a single KL, a preserved CEB certificate, and prediction rules tied to the geometry (the anchor-energy classifier and the projection head); (3) a proof that the CEB certificate is blind to inter-label geometry, the removal of the unspecified inter-class geometry of the CEB reference (adding explicit geometric constraints to the reference distribution), and a unified framework theory — certificate preservation and conditional separation transfer for any qualified geometric prior family, upgraded by β-controlled anchor-alignment bounds at near-optima in the deterministic K-class setting and in the deterministic regression setting (with $C_{\mathrm{align}} = \tau^2/\rho^2$ for the isometric instance) — instantiated by the orthogonal frame and the isometric axis, together with a mechanism connection to the Caveats fix of the dead knob (the anchored reference supplies the quadratic compression term that squared-IB installs by hand); (4) a mechanism-level empirical validation of the orthogonal instance: the constructed geometry is transferred to the posterior and is sufficient for prediction.

2 Method

2.1 Background and setup

Let X be the input and Y the label with K classes; the encoder induces the Markov chain $Y - X - Z$. Following Fischer [2020], the variational CEB objective is

$$\mathrm{VCEB} = \min_{e,b,c} \big\langle \log e(z|x) \big\rangle - \big\langle \log b(z|y) \big\rangle - \gamma \big\langle \log c(y|z) \big\rangle, \tag{1}$$

where $e(z|x)$ is the stochastic encoder, $b(z|y)$ the backward encoder, $c(y|z)$ the classifier, and the expectations are over $p(x,y)\, e(z|x)$. We write the encoder as the diagonal Gaussian posterior $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \mathrm{diag}(\sigma^2_\phi(x)))$ and use the loss convention $\mathrm{CE} + \beta\ \mathrm{KL}$, with $\beta = 1/\gamma$; at evaluation the deterministic code $z = \mu_\phi(x)$ is used. The central object is the residual certificate: for any conditional prior $r(z|y)$,

$$\mathbb{E}\big[\mathrm{KL}(q_\phi(z|x) \,\|\, r(z|y))\big] = I(X;Z|Y) + \mathbb{E}_y\big[\mathrm{KL}(q_\phi(z|y) \,\|\, r(z|y))\big] \ \geq\ I(X;Z|Y), \tag{2}$$

by Fischer's chain-rule identity (their Eqs. 4 and 9–11, 20). CEB instantiates $r(z|y) = b_\theta(z|y)$ with a trainable linear map of the one-hot label; VIB fixes $r(z|y) = \mathcal{N}(0, I)$ for all y. Our framework replaces r with a geometric prior family, defined in Section 2.2.4.

2.2 Key theory: the failure channels of CEB and the guarantees of a geometric prior

2.2.1 Channel 1 (dead knob): the Caveats fix and the anchored-quadratic mechanism

Proposition 2.1 (Dead knob, following Caveat 1 of Kolchinsky and Tracey [2019]). *Assume $Y = f(X)$. For every $\gamma > 0$, the CEB objective $L_\gamma(Z) := I(X; Z|Y) - \gamma I(Y; Z)$ is minimized exactly on the corner family $\{Z : I(Y; Z) = H(Y),\ I(X; Z|Y) = 0\}$, and the map $\gamma \mapsto \big(I(X; Z^*|Y),\ I(Y; Z^*)\big)$ is constant. Consequently no interior point of the IB curve is ever a Lagrangian minimizer of CEB, over the entire parameter range.*

Proof sketch (see Kolchinsky and Tracey [2019] for the full argument). By the chain rule, $L_\gamma(Z) = I(X; Z) - (1 + \gamma)I(Y; Z)$; the data-processing inequality gives $L_\gamma \geq -\gamma H(Y)$, attained exactly at the corner $Z = Y$. In the Caveats paper's maximizing convention this is their Eq. 7 with $\beta' = 1/(1 + \gamma) \in (0, 1)$, the range in which every maximizer is pinned at the copy corner. □

The corner is a single point of the information plane and an entire equivalence class of representations; the objective is exactly indifferent within the class, and neither the objective nor the certificate records which member the optimizer reaches. In the deterministic scenario this corner *is* the MNI point, so CEB's declared target and the degenerate attractor coincide; the γ knob acts on the optimization trajectory, not on the optimum. The Caveats paper repairs this by squaring the compression term. Our geometric prior supplies the same strict convexity by a different mechanism, without squaring: on the aligned stochastic surrogate, with the main model's anchor-energy classifier fixing the classifier scale by construction, the training Lagrangian is strictly convex jointly in the alignment and scale variables, with a unique interior optimum per β (for $a > 0$) (Proposition A.1, Appendix; Proposition 2.10) — the anchored reference contributes a quadratic displacement term by construction, playing the role of the squared compression term. We present this as a mechanism connection to the Caveats fix, not as a new solution of the deterministic degeneration: the analysis is confined to the aligned surrogate, whether the optimum moves monotonically in β is not established, and the β-sweep is reported empirically in the experiments.

2.2.2 Channel 2 (blind certificate): the certificate is blind to the relationships between labels

Proposition 2.2 (The certificate records nothing about inter-label geometry). *For any* fixed *encoder, with the backward encoder matched to the posterior per class ($b = q(z|y)$), the certificate collapses to the residual,*

$$\mathrm{ReX} = I(X; Z|Y), \tag{3}$$

whatever the pairwise geometry of the class-conditional priors $\{q(z|y)\}$: class means merged, coincident, or placed in arbitrary directions all yield the same certificate value. Formally, the certificate *contains no cross-class term — it is a sum of per-class* *matching terms, each sample's KL touches only its own class prior, and the matching condition can be met class by class regardless of the geometry of* $\{q(z|y)\}$ *— so the certificate never records the relationships among the class priors. (The prediction term of the full objective couples the classes through the softmax; we make no claim about the full objective here, which is why the proposition is stated at a fixed encoder with a matched backward encoder.)*

Proof. For any matched pair ($b = q(z|y)$), the gap identity of Eq. (2) gives Eq. (3); the matching condition is stated per class and no term of the certificate couples two distinct class priors, so the certificate value carries no information about the pairwise geometry of $\{q(z|y)\}$. Note that $I(X; Z|Y)$ is generally non-zero; it vanishes only under the conditional independence $Z \perp X \mid Y$, which is a property of the encoder, not of the prior. $\square$

Proposition 2.3 (Retention blindness on T_α)**.** *On the manifold* $T_\alpha = B_\alpha \cdot Y$ *of Kolchinsky and Tracey [2019] (their Eq. 6) — a family that depends on* Y *alone —* $I(X; T_\alpha|Y) = 0$ *for every* α*, and with the optimal backward encoder and classifier,* $\mathrm{ReX}(T_\alpha) = 0$ *and* $\mathrm{CE}(T_\alpha) = H(Y) - I(Y; T_\alpha) =: \delta_\alpha$*: the certificate reads* 0 *from the total collapse (*$\alpha = 0$*) to the full copy (*$\alpha = 1$*), while the retained label information* $I(Y; T_\alpha)$ *sweeps the whole axis. Tightness of the certificate is therefore compatible with retaining all, part, or none of* Y *— the certificate records nothing about how much label information the code retains.*

Proof. T_α is a function of Y alone, hence $T_\alpha \perp X \mid Y$ and $I(X; T_\alpha|Y) = 0$. By Proposition 2.2 with the true backward encoder the certificate is tight, $\mathrm{ReX} = I(X; T_\alpha|Y) = 0$, and the optimal classifier attains $\langle \log c(y|z) \rangle = -H(Y|T_\alpha)$, i.e. $\mathrm{CE} = H(Y|T_\alpha) = \delta_\alpha$. $\square$

Remark 2.1 (The objective on T_α is β-independent and strictly prefers the full copy). On T_α the compression term contributes nothing ($\mathrm{ReX} \equiv 0$), so the objective reads $L = \mathrm{CE} + \beta\, \mathrm{ReX} = \delta_\alpha$ — independent of β, and strictly minimized at $\alpha = 1$ (the full copy). No certified residual is produced anywhere on the manifold; the blindness is that the certificate cannot tell the optimizer — or the analyst — where on the retention axis the code sits. Any statement about trading label information against certified residual would require a path on which $\mathrm{ReX} > 0$; on T_α no such trade occurs.

The two channels locate the disease precisely: the dead knob inherits the linear degeneracy of the deterministic IB curve (Channel 1, cited), and the certificate contains no cross-class term, so it is blind to the relationships among the class priors (Channel 2, proven above). The construction below *removes the unspecified inter-class geometry of the CEB reference*: the prior declares the relationships between labels by construction, and the certificate becomes a distance to that declared geometry — a representation whose class geometry differs from the declared one pays a positive certificate (Propositions 2.9 and 2.3). The framework does not claim to make the certificate read off the retained label information.

2.2.3 The GPB framework and the certificate

The framework is a single-parameter family of objectives. Let $\mathcal{G}$ be a *qualified geometric prior family*: a set of label-conditional distributions $p_\theta(z|y)$ that depend on y alone, computed from the prior encoder, and satisfying two checkable conditions — a bounded, diagonal covariance (the diagonal variational family of the VIB/CEB lineage),

$$p_\theta(z|y) = \mathcal{N}\big(m_\theta(y),\, \Sigma_\theta(y)\big), \qquad \Sigma_\theta(y) \text{ diagonal}, \qquad \Sigma_\theta(y) \;\preceq\; \tau_{\max}^2\, I_d \text{ for all } y, \tag{4}$$

and a mean map that encodes the label structure at a fixed, declared scale, in one of two forms. For *discrete* labels, a two-sided separation condition,

$$\Delta \;\leq\; \|m_\theta(y) - m_\theta(y')\| \;\leq\; \bar{\Delta} \text{ for all } y \neq y', \tag{5}$$

for *continuous* labels with a declared label metric d_Y, a two-sided metric condition,

$$\rho_{\min}\, d_Y(y, y') \;\leq\; \|m_\theta(y) - m_\theta(y')\| \;\leq\; \rho_{\max}\, d_Y(y, y'), \tag{6}$$

where $\Delta, \bar{\Delta}, \rho_{\min}, \rho_{\max}$ are declared constants of the family, fixed before training and not trained. The lower bounds encode the label distinctions (the pairwise continuous form imposes no global minimum separation, which is necessarily zero on a continuous label space); the upper bounds close the scale escape. Without them, the family would contain $m_\theta(y) = s\, m_0(y)$ for arbitrarily large s — with the posterior following, the KL could stay zero while the prediction term improves — so the objective might lack a finite optimum and "geometry by construction" would hold only in a vacuous sense. The objective is

$$\mathcal{L}_{\mathrm{GPB}} = \mathbb{E}_{(x,y)}\Big[\mathrm{CE}\big(c_\psi(z), y\big) + \beta\ \mathrm{KL}\big(q_\phi(z|x) \,\|\, p_\theta(z|y)\big)\Big], \qquad z \sim q_\phi(z|x), \qquad p_\theta \in \mathcal{G}. \tag{7}$$

Proposition 2.4 (A stateless geometric prior preserves the CEB certificate)**.** *For any prior family $\mathcal{G}$ whose members are conditional priors independent of x, and for every parameter value θ at which $p_\theta \in \mathcal{G}$ is defined, the gap identity of Eq.* (2) *applies pointwise in θ:*

$$\mathbb{E}\Big[\mathrm{KL}\big(q_\phi(z|x) \,\|\, p_\theta(z|y)\big)\Big] = I_{q_\phi}(X; Z|Y) + \mathbb{E}_Y\Big[\mathrm{KL}\big(q_\phi(z|Y) \,\|\, p_\theta(z|Y)\big)\Big] \;\geq\; I_{q_\phi}(X; Z|Y). \tag{8}$$

The recomputation of p_θ after parameter updates does not invalidate this identity: at each fixed θ it remains a function of y only. If the prior were computed from the current input batch, this conditional-prior argument would no longer apply without a new analysis.

Proof. For fixed θ, apply the gap identity of Eq. (2) to the conditional distribution $p_\theta(z|y)$ and the posterior q_ϕ. □

Proposition 2.5 (Separation transfer for any qualified geometric prior)**.** *Let $\mathcal{G}$ be qualified with variance bound $\tau^2_{\max}$.* (a) Discrete labels. *Let $m_k := \mathbb{E}[\mu_\phi(X)|Y = k]$ be the posterior class centers and define the class-wise anchor mismatch*

$$D_k := \frac{1}{2\tau^2_{\max}} \mathbb{E}\Big[\|\mu_\phi(X) - m_\theta(k)\|^2 \,\Big|\, Y = k\Big]. \tag{9}$$

Then for every pair $i \neq j$,

$$\|m_i - m_j\| \;\geq\; \|m_\theta(i) - m_\theta(j)\| - \sqrt{2\tau^2_{\max} D_i} - \sqrt{2\tau^2_{\max} D_j} \;\geq\; \Delta - \sqrt{2\tau^2_{\max} D_i} - \sqrt{2\tau^2_{\max} D_j}, \tag{10}$$

using Eq. (5)*.* (b) Continuous labels. *Assume X and Y are Borel-measurable random variables (e.g., Y real-valued), so that regular conditional distributions exist, and $\mu_\phi(X)$ is integrable. Define the regular conditional posterior center and the pointwise anchor mismatch,*

$$m(y) := \mathbb{E}\big[\mu_\phi(X) \mid Y = y\big], \qquad D(y) := \frac{1}{2\tau^2_{\max}} \mathbb{E}\Big[\|\mu_\phi(X) - m_\theta(y)\|^2 \,\Big|\, Y = y\Big], \tag{11}$$

both well-defined for P_Y-almost every y. Then for P_Y-almost every pair $y \neq y'$,

$$\|m(y) - m(y')\| \;\geq\; \|m_\theta(y) - m_\theta(y')\| - \sqrt{2\tau^2_{\max} D(y)} - \sqrt{2\tau^2_{\max} D(y')} \;\geq\; \rho_{\min}\, d_Y(y, y') - \sqrt{2\tau^2_{\max} D(y)} - \sqrt{2\tau^2_{\max} D} \tag{12}$$

using Eq. (6)*; no global positive minimum separation is required. Small pointwise anchor mismatch therefore guarantees that the posterior centers inherit the prior's separation for* any *qualified member of $\mathcal{G}$ — this is the unified, geometry-independent content of the framework.*

Proof. For (a), Jensen gives $\|m_k - m_\theta(k)\| \leq \sqrt{2\tau^2_{\max} D_k}$, and the triangle inequality on $m_i - m_j = (m_i - m_\theta(i)) + (m_\theta(i) - m_\theta(j)) + (m_\theta(j) - m_j)$ gives Eq. (10). For (b), the same two inequalities hold pointwise for P_Y-a.e. y: Jensen on the regular conditional expectation gives $\|m(y) - m_\theta(y)\| \leq \sqrt{2\tau^2_{\max} D(y)}$ (the norm is convex), and the triangle inequality gives Eq. (12); intersecting the two measure-one sets of valid y's yields the almost-everywhere pair statement. □

Remark 2.2 (Finite-sample estimation of the continuous case). On finite samples the regular conditional expectations of part (b) are estimated by binning the label axis into intervals B_i with representative labels $\bar{y}_i$ and centers $m_i = \mathbb{E}[\mu_\phi(X) \mid Y \in B_i]$; the bin diameter adds a quantization error bounded by $\rho_{\max}\,\mathrm{diam}(B_i) + \rho_{\max}\,\mathrm{diam}(B_j)$ (by the upper bound of Eq. (6)), which vanishes as the bins shrink. The theorem itself is stated pointwise and needs no binning.

Proposition 2.5 is conditional on the anchor mismatch being small. The next result supplies the missing link: it shows that minimizing the objective itself forces the expected mismatch below an explicit β-dependent bound.

Proposition 2.6 (Anchor alignment at near-optima: deterministic K-class classification)**.** *Assume the deterministic K-class classification setting: $Y = f(X)$, a cross-entropy prediction loss, and a qualified family whose diagonal covariances belong to* *the diagonal family the posterior can represent. Suppose further that the encoder family contains class-constant maps (standard in the overparameterized setting), so that the aligned solution $q(z|x) = p_\theta(z|y)$ is feasible, and that the classifier family contains the Bayes posterior of the aligned mixture,*

$$c_k(z) = \frac{\pi_k\,|\Sigma_\theta(k)|^{-1/2}\,e^{-\frac{1}{2}(z-m_\theta(k))^\top\Sigma_\theta(k)^{-1}(z-m_\theta(k))}}{\sum_j \pi_j\,|\Sigma_\theta(j)|^{-1/2}\,e^{-\frac{1}{2}(z-m_\theta(j))^\top\Sigma_\theta(j)^{-1}(z-m_\theta(j))}}, \tag{13}$$

including the Gaussian normalization factors $|\Sigma_\theta(k)|^{-1/2}$. For the orthogonal instance, where all classes share the covariance $\tau^2 I_d$ and the means are the anchors $a\,q_{\theta,k}$, the determinant factors cancel and this Bayes posterior reduces to exactly the anchor-energy classifier of Eq. (27)*, whose weights $w_k = (a/\tau^2)\,q_k$ are fixed by construction. Then for any solution (ϕ,θ,ψ) with objective value $L \le L^* + \varepsilon$, where L^* is the infimum of Eq.* (7)*, the expected anchor mismatch is controlled by β:*

$$\mathbb{E}\big[D(Y)\big] \;\le\; \frac{C_{\mathrm{G}}+\varepsilon}{\beta} \;\le\; \frac{\ln K+\varepsilon}{\beta}, \qquad D(y) := \frac{1}{2\tau_{\max}^2}\,\mathbb{E}\Big[\|\mu_\phi(X) - m_\theta(y)\|^2 \;\Big|\; Y = y\Big], \tag{14}$$

where $C_{\mathrm{G}} := H(Y \mid Z_{\mathrm{align}})$ is the cross-entropy of the Bayes classifier of Eq. (13) *on the aligned mixture, bounded by $C_{\mathrm{G}} \le H(Y) \le \ln K$; in the shared-covariance orthogonal instance it specializes to the anchor-energy classifier's aligned cross-entropy C_{OPB}. For tasks in which no zero-KL feasible solution exists — in particular for continuous regression, where the encoder cannot recover y from x, and where no such Bayes comparison classifier is defined — the argument does not apply: there the separation transfer of Proposition 2.5 remains a conditional statement on the anchor mismatch, and no framework-level alignment bound is claimed.*

Proof. Since $\Sigma_\theta(y) \preceq \tau_{\max}^2 I_d$, each sample's KL contains a mean-mismatch term with variance at most $\tau_{\max}^2$, so $\mathrm{KL}(q_\phi(z|x)\,\|\,p_\theta(z|y)) \ge \|\mu_\phi(x) - m_\theta(y)\|^2/(2\tau_{\max}^2)$, and taking expectations gives $\mathbb{E}[\mathrm{KL}] \ge \mathbb{E}[D(Y)]$. The comparison solution $q(z|x) = p_\theta(z|y)$ is feasible by the assumptions ($Y = f(X)$, class-constant maps, diagonal representable covariances), has zero KL, and its prediction is the Bayes posterior of Eq. (13), whose cross-entropy is exactly the conditional entropy $C_{\mathrm{G}} = H(Y \mid Z_{\mathrm{align}}) \le \ln K$; for the orthogonal instance the anchor-energy classifier of the main model realizes it (shared covariance, determinant factors cancel). Hence $L^* \le C_{\mathrm{G}}$ and $\beta\,\mathbb{E}[\mathrm{KL}] \le L \le L^* + \varepsilon \le C_{\mathrm{G}} + \varepsilon \le \ln K + \varepsilon$, whence Eq. (14). □

Corollary 2.1 (Separation at near-optimal solutions, classification)**.** *Under the assumptions of Proposition 2.6, for classes $i \ne j$ with $\pi_i, \pi_j > 0$,*

$$\|m_i - m_j\| \;\ge\; \|m_\theta(i) - m_\theta(j)\| - \sqrt{\frac{2(C_{\mathrm{G}}+\varepsilon)}{\beta\,\pi_i}} - \sqrt{\frac{2(C_{\mathrm{G}}+\varepsilon)}{\beta\,\pi_j}}, \tag{15}$$

which specializes via Δ (Eq. (5)*). In particular, at a global optimum ($\varepsilon = 0$), classes with probability at least $\pi_{\min}$ have posterior centers separated by at least $\Delta - 2\sqrt{2C_{\mathrm{G}}/(\beta\,\pi_{\min})}$ (with $C_{\mathrm{G}} \le \ln K$; in the orthogonal instance $C_{\mathrm{G}} = C_{\mathrm{OPB}}$, the anchor-energy classifier's aligned cross-entropy).*

Proof. $D_k \le \mathbb{E}[D(Y)]/\pi_k \le (C_\mathrm{G}+\varepsilon)/(\beta\pi_k)$; combine with Proposition 2.5, whose mismatch terms are $\sqrt{2\tau_{\max}^2 D_k}$. □

Proposition 2.7 (Anchor alignment at near-optima: deterministic continuous regression)**.** *Assume the deterministic regression setting $Y = f(X)$ with a non-negative regression loss $\ell(\hat{y}, y)$ (e.g., the squared loss on standardized labels), a qualified family, a posterior able to realize the aligned solution $q(z|x) = p_\theta(z|y)$ (the encoder family contains the corresponding maps), and the tied projection head of the isometric instance. Let C_{align} be the regression risk of the ideal aligned solution — for the standardized squared loss and the EPB head, $\hat{\tilde{y}} = u^\top z/\rho$ with $z \sim p_\theta(z|y)$ gives $\hat{\tilde{y}} = \tilde{y} + (\tau/\rho)\,\eta$ with $\eta \sim \mathcal{N}(0,1)$, so $C_{\mathrm{align}} = \tau^2/\rho^2$. Then for any solution with objective value $L \le L^* + \varepsilon$,*

$$\mathbb{E}\big[D(Y)\big] \;\le\; \frac{C_{\mathrm{align}}+\varepsilon}{\beta}, \tag{16}$$

with $D(y)$ as in Eq. (11).

Proof. As in Proposition 2.6, $\mathbb{E}[\mathrm{KL}] \ge \mathbb{E}[D(Y)]$ by the covariance bound. The comparison solution is feasible by the assumptions ($Y = f(X)$, aligned maps representable): its posterior is $q(z|x) = p_\theta(z|y)$ (zero KL) and its prediction is the tied head, whose risk is $C_{\mathrm{align}} = \tau^2/\rho^2$ (the prediction error is exactly the Gaussian bottleneck noise $(\tau/\rho)\,\eta$). Hence $L^* \le C_{\mathrm{align}}$ and $\beta\,\mathbb{E}[\mathrm{KL}] \le L \le L^* + \varepsilon \le C_{\mathrm{align}} + \varepsilon$, whence Eq. (16). □

Corollary 2.2 (Geometric guarantee on high-probability label sets, regression)**.** *Under Proposition 2.7, for any $t > 0$, Markov's inequality gives $P_Y\big(D(Y) > t\big) \le (C_{\mathrm{align}} + \varepsilon)/(\beta t)$, and on the label set $S_t := \{y : D(y) \le t\}$, of P_Y-measure at least $1 - (C_{\mathrm{align}}+\varepsilon)/(\beta t)$, the pointwise transfer of Proposition 2.5(b) yields, for all $y, y' \in S_t$,*

$$\|m(y) - m(y')\| \;\ge\; \rho_{\min}\, d_Y(y, y') - 2\sqrt{2\tau_{\max}^2\, t}. \tag{17}$$

For EPB ($\rho_{\min} = \rho$, $\tau_{\max} = \tau$) the bound reads $\rho\, d_Y(y, y') - 2\sqrt{2\tau^2 t}$: the posterior geometry is controlled by β, ρ, and the bottleneck noise τ — the optimization-level guarantee for the regression instance.

The framework therefore has two geometry-independent statements — certificate preservation (Proposition 2.4) and the conditional separation transfer (Proposition 2.5); additionally, the β-controlled anchor alignment at near-optima upgrades the transfer into an optimization guarantee in the deterministic K-class classification setting (Proposition 2.6; Corollary 2.1) and in the deterministic continuous regression setting (Proposition 2.7; Corollary 2.2). Everything below is a property of a specific instance satisfying Eqs. (4)–(6).

2.2.4 The orthogonal-frame instance (OPB)

For classification, let $K \le d$. The prior encoder g_θ is a single unactivated linear map from one-hot labels to $\mathbb{R}^d$, evaluated on *all* K class one-hot vectors at every forward pass. Writing $E_K = I_K$ for the all-class one-hot matrix, the encoder output is the raw class-mean matrix

$$U_\theta = g_\theta(E_K)^\top \in \mathbb{R}^{d\times K}. \tag{18}$$

The frame is the thin polar factor of U_θ,

$$Q_\theta = U_\theta\big(U_\theta^\top U_\theta\big)^{-1/2}, \qquad Q_\theta^\top Q_\theta = I_K, \tag{19}$$

the matrix square root being computed via the eigendecomposition of $U_\theta^\top U_\theta$, or equivalently via the thin SVD $U_\theta = W\Sigma V^\top$ with $Q_\theta = WV^\top$. The polar factor is the unique closest orthonormal matrix to U_θ, needs no sign convention, and is exactly equivariant under relabeling of the classes: for any label permutation matrix P,

$$Q_\theta(U_\theta P) = U_\theta P\big(P^\top U_\theta^\top U_\theta P\big)^{-1/2} = U_\theta\big(U_\theta^\top U_\theta\big)^{-1/2} P = Q_\theta(U_\theta)\, P, \tag{20}$$

since $PP^\top = I_K$ and the inverse square root conjugates under P. Permuting the label numbering therefore permutes the anchors accordingly, and no class index plays a distinguished role. We write $q_{\theta,k} := Q_\theta[:,k]$ and make the following standing assumption, verified numerically in the implementation (Section 2.3.1).

Assumption 2.1 (Full rank)**.** *U_θ has full column rank at every training step, so that Eq.* (19) *— and its backward pass — is defined. The condition holds almost surely at the identity initialization below; the implementation* verifies *it by monitoring $\sigma_{\min}(U_\theta)$ along training (Section 2.3.1), and no additional regularizer is imposed — the objective remains Eq.* (23) *exactly.*

The geometric prior replaces the raw means by the scaled orthogonal frame

$$p_\theta^{\mathrm{OPB}}(z|y=k) = \mathcal{N}\big(a\, q_{\theta,k},\, \tau^2 I_d\big), \tag{21}$$

where $a > 0$ is the anchor scale and $\tau^2 > 0$ the *fixed shared* prior variance (default 1.0). The variance is deliberately not learnable: a learnable per-class variance would give the reference an extra channel to absorb the compression cost by inflating its covariance — precisely the degeneracy the geometric construction excludes — and with τ^2 fixed, the guarantees of Propositions 2.9–2.3 hold for the actual optimization problem. The frame $\widehat{Q}_\theta$ is an immediate function of the current prior-encoder parameters, computed from the all-class table: there is no EMA prototype, no persistent anchor variable, and, for fixed θ, $\widehat{Q}_\theta$ depends only on the class set — not on the input x, and not on the batch.

Proposition 2.8 (Orthogonal class-prior geometry by construction)**.** *Under Eqs.* (19)*–*(21)*, the class means $a_k = a\, q_{\theta,k}$ satisfy*

$$a_i^\top a_j = a^2\delta_{ij}, \qquad \|a_k\| = a, \qquad \|a_i - a_j\|^2 = 2a^2 \;\; (i \neq j). \tag{22}$$

Equivalently, the class-mean Gram matrix is $a^2 I_K$: under Assumption 2.1, the prior supplies a class-separated spherical code at every training step, whatever the optimizer does.

Proof. Eq. (22) follows from $Q_\theta^\top Q_\theta = I_K$ after scaling by a; in particular $\|a_i - a_j\|^2 = \|a_i\|^2 + \|a_j\|^2 - 2a_i^\top a_j = 2a^2$. □

The single objective of the variant is

$$\mathcal{L}_{\mathrm{OPB}} = \mathbb{E}_{(x,y)}\Big[\mathrm{CE}\big(c_\psi(z), y\big) + \beta\ \mathrm{KL}\big(q_\phi(z|x)\,\|\,p_\theta^{\mathrm{OPB}}(z|y)\big)\Big], \qquad z \sim q_\phi(z|x). \tag{23}$$

The defining change from CEB is the prior of Eq. (21), not a second latent variable or a second KL term.

2.2.5 Instance guarantees: a constructive class geometry and conditional center separation

Proposition 2.9 (Positive rate cost of class-independent means)**.** *Let $\pi_k = \Pr(Y = k)$ and suppose the posterior mean is class-independent, $\mathbb{E}[\mu_\phi(X)|Y = k] = c$ for every k. Then the mean-mismatch part of the rate satisfies*

$$\mathbb{E}\Big[\tfrac{1}{2\tau^2}\|\mu_\phi(X) - a\,q_{\theta,Y}\|^2\Big] \ \geq\ \frac{a^2}{2\tau^2}\Big(1 - \sum_{k=1}^{K}\pi_k^2\Big), \tag{24}$$

with equality at the optimal constant mean $c = a\sum_k \pi_k q_{\theta,k}$. For uniform classes the bound is $a^2(1 - 1/K)/(2\tau^2)$. Unlike a zero-mean marginal prior, the geometric prior assigns every class-independent posterior a strictly positive mean-matching cost whenever $a > 0$ and two classes have positive probability.

Proof. By Jensen's inequality the left side is at least $\frac{1}{2\tau^2}\sum_k \pi_k\|c - aq_{\theta,k}\|^2 = \frac{1}{2\tau^2}\big(\|c\|^2 - 2a\,c^\top\sum_k \pi_k q_{\theta,k} + a^2\big)$. Minimizing over c gives $c = a\sum_k \pi_k q_{\theta,k}$; by orthonormality $\|\sum_k \pi_k q_{\theta,k}\|^2 = \sum_k \pi_k^2$, yielding Eq. (24). $\square$

Corollary 2.3 (The orthogonal instance satisfies the framework separation transfer)**.** *The orthogonal instance is qualified with $\tau^2_{\max} = \tau^2$ and $\Delta = \bar{\Delta} = \sqrt{2}\,a$ (all pairwise anchor distances equal $\sqrt{2}\,a$ by Eq. (22), with the anchor scale a fixed); Proposition 2.5 therefore specializes to*

$$\|m_i - m_j\| \ \geq\ \sqrt{2}\,a - \sqrt{2\tau^2 D_i} - \sqrt{2\tau^2 D_j}, \tag{25}$$

with the class-wise anchor mismatch

$$D_k := \frac{1}{2\tau^2}\,\mathbb{E}\Big[\|\mu_\phi(X) - a\,q_{\theta,k}\|^2\ \Big|\ Y = k\Big]. \tag{26}$$

Proposition 2.10 (Anchor-energy classifier equivalence)**.** *Suppose the classifier uses the anchor energy*

$$s_k(z) = -\frac{\|z - a\,q_{\theta,k}\|^2}{2\tau^2} + \log\pi_k. \tag{27}$$

Then $s_k(z)$ is the class-k Gaussian log posterior up to a class-independent term; equivalently, for uniform classes the equivalent linear classifier has weights $w_k = (a/\tau^2)\,q_{\theta,k}$ — equal norms and pairwise orthogonal rows. An anchor-energy classifier therefore makes the prediction rule use exactly the same geometry as the prior, and removes the free classifier scale by construction: its scale is fixed to a/τ^2, with no independent margin parameter — the condition under which Proposition A.1 holds for the actual training objective.

Proof. Expand the squared norm in Eq. (27); the terms $-\|z\|^2/(2\tau^2)$ and $-a^2/(2\tau^2)$ are common to all classes, and the class-dependent remainder is the Gaussian discriminant score. $\square$

Remark 2.3 (What the framework constrains — and what it does not). The constraints of the geometric prior bind the reference, not the deployed representation: for any orthonormal frame $\{q'_k\}$ and the class-constant posterior $q(z|Y=k) = \mathcal{N}(a\,q'_k, \tau^2 I_d)$, the trainable frame can rotate to match ($q_{\theta,k} \to q'_k$), yielding $\mathrm{KL} = 0$, $I(X;Z|Y) = 0$, and $I(Y;Z) > 0$ — a full-retention label code with a zero certificate. The framework therefore constrains the geometric *form* of the label code — precisely what CEB left unspecified — and does not claim that the certificate meters retained label information. (With the variance fixed, the remaining freedom at certificate zero is the frame rotation; a learnable per-class variance would add a further circumvention channel and is deliberately not used.)

Propositions 2.4–2.3 are the core of the theory. The remaining results — the pointwise KL decomposition, a Fano-type label-information lower bound, the strict convexity of the β-sweep, and the scope and limitations of the framework — are given in the appendix.

2.3 Model variants

The framework has two concrete instantiations, one per label-space structure. Both keep the CEB posterior, the single-KL objective, and the residual certificate unchanged; they differ only in the geometric family that constrains the prior.

2.3.1 OPB: an orthogonal class frame (classification)

Posterior. The encoder computes $h = f_{\mathrm{enc}}(x)$; two heads produce the diagonal posterior $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \mathrm{diag}(\sigma_\phi^2(x)))$; the code z is sampled at training and set to $z = \mu_\phi(x)$ at evaluation.

Classifier. The main model uses the anchor-energy classifier of Proposition 2.10 (Eq. (27)): the prediction rule and the prior share one geometry, and the classifier has *no free parameters* — its scale is fixed to a/τ^2 by construction, which is what allows the strict-convexity analysis of Proposition A.1 to be carried out on the training objective itself (restricted to the aligned family). The plain linear classifier is kept as a *free-classifier variant* in the experiments; there the classifier scale is a free parameter, and the analysis of Proposition A.1 does not cover it (an analysis of the free classifier would require an explicit norm constraint or regularizer).

Geometric prior. The prior encoder is a single unactivated linear layer $g_\theta : \mathbb{R}^K \to \mathbb{R}^d$; at every forward pass the *full* all-class one-hot matrix E_K is passed through it, and the resulting raw class-mean matrix U_θ of Eq. (18) is orthonormalized by the thin polar factor of Eq. (19) (computed via the thin SVD $U_\theta = W\Sigma V^\top$, $Q_\theta = WV^\top$; no sign convention is needed). The prior is the scaled frame of Eq. (21) with the fixed shared variance τ^2. The loss is Eq. (23): a single KL.

Statelessness. Q_θ is an immediate function of the current prior encoder: the next step recomputes it. There is no EMA, no prototype table, and no dependence on the batch composition — the property that makes Proposition 2.4 exact.

Initialization. The posterior logvar head is zero-initialized ($\sigma^2 = \tau^2 = 1$ at initialization, so training starts at KL $\approx \frac{1}{2\tau^2}a^2$); the prior encoder is identity-initialized, so the initial raw matrix is $U_\theta = [e_1, \ldots, e_K]$ — full column rank, with a stable polar-factor backward (the verified condition of Assumption 2.1).

Theory–code layering. The prior variance is the fixed shared value τ^2 (default 1.0) in both the theory and the released implementation: a learnable per-class variance is deliberately not used, as it would give the reference an extra channel to absorb the compression cost by inflating its covariance — the very degeneracy the geometric construction excludes (see also Remark 2.3). The released implementation uses exactly the polar prior encoder of Eqs. (18)–(19). Orthonormality holds exactly wherever the factorization is defined; the full-rank condition of Assumption 2.1 is *verified, not imposed*: $\sigma_{\min}(U_\theta)$ is monitored along training and stays bounded away from zero in the reported runs, and no additional regularizer is added to the objective, which remains Eq. (23) exactly. Near rank deficiency the polar factor's gradient is delicate; should the monitor flag a near-deficient step, the recommended handling is a re-initialization of the prior encoder (the reported runs never triggered it).

2.3.2 EPB: an equidistant regression axis (regression)

Setup. For a one-dimensional continuous label, standardize once on the training set: $\tilde{y} = (y - y_{\text{center}})/y_{\text{scale}}$, with the statistics frozen for validation and test. Let $W \in \mathbb{R}^{d\times 1}$ be a trainable direction and $u = W/\|W\|$ its per-step normalization. The isometric prior is

$$p^{\text{EPB}}(z|y) = \mathcal{N}\big(\rho\,\tilde{y}\,u,\ \tau^2 I_d\big), \qquad \|\mu_p(y_i) - \mu_p(y_j)\| = \rho\,|\tilde{y}_i - \tilde{y}_j|, \tag{28}$$

where $\rho > 0$ is the isometric scale and $\tau^2 > 0$ the fixed shared prior variance, as in the classification variant (the isometry statement involves the means only and is unaffected by the variance). EPB is the isometric instance of the framework: it satisfies Eq. (4) with $\tau^2_{\max} = \tau^2$ and the two-sided metric condition of Eq. (6) with the standardized label metric and $\rho_{\min} = \rho_{\max} = \rho$ — in fact with equality on both sides, by construction, so the isometry pins the scale. The pointwise separation transfer of Proposition 2.5(b) therefore applies to its posterior centers, and, in the deterministic regression setting, the alignment bound of Proposition 2.7 applies with $C_{\text{align}} = \tau^2/\rho^2$. The isometry is exact and global: any nonzero linear map of a scalar label is isometric up to its norm, and the normalization fixes that norm to ρ — unlike fixed random-feature anchors, whose pairwise distances grow nonlinearly and saturate, the isometric axis preserves label distances at every scale (with the price that anchor norms grow with $|\tilde{y}|$; a line and a fixed-radius sphere intersect in at most two points, so isometry and equal norms cannot both hold). All prior means lie on one line; for $y_i < y_j < y_k$ the distances add, $\|\mu_p(y_i) - \mu_p(y_k)\| = \|\mu_p(y_i) - \mu_p(y_j)\| + \|\mu_p(y_j) - \mu_p(y_k)\|$: order and magnitude of label differences are encoded exactly.

Axis choices. Three variants of the direction: (A) *fixed axis*, $u = [1, 0, \ldots, 0]$, with no trainable prior direction — the theoretically cleanest ablation, at the price of prespecifying a latent coordinate; (B) *trainable normalized axis*, $u = W/\|W\|$ with W trained and normalized at every step — our main variant, which keeps the scale ρ exact while allowing the model to choose the axis; (C) *unconstrained linear axis*, $\mu_p(y) = W\tilde{y}$ — the simplest ablation, still isometric by linearity but with a free scale that can shrink and weaken the geometry. We recommend (B), use (A) to test whether the trainable rotation carries the benefit, and keep (C) as a diagnostic. The main variant uses no bias: an affine offset cancels in pairwise distances but is a free extra degree of matching, and $\tilde{y}$ is already centered.

Prediction head. The main model uses the *tied projection head*, which predicts on the same axis the KL uses, $\hat{\tilde{y}} = u^\top z/\rho$, then rescales $\hat{y} = y_{\text{center}} + y_{\text{scale}}\,\hat{\tilde{y}}$ — the regression analogue of the anchor-energy classifier: geometrically consistent, with no free direction or scale that could bypass the prior. The *ordinary regression head* $\hat{y} = \text{regressor}(z)$ is kept as a free-head variant for a fair comparison against CEB. The total objective is $\mathcal{L}_{\text{EPB}} = L_{\text{reg}}(\hat{y}, y) + \beta\ \text{KL}\big(q_\phi(z|x)\,\|\,p^{\text{EPB}}(z|y)\big)$ — a single KL, no EMA, no label bucketing, no batch-level orthonormalization.

Relation to the classification variant. OPB expresses the discrete structure of the label space — every pair of distinct classes at the same distance $\sqrt{2}a$; EPB expresses the continuous structure — latent distance proportional to label distance, $\rho|\Delta\tilde{y}|$. Both replace CEB's unconstrained reference with a geometry that encodes the label space by construction, and both preserve the posterior, the single KL, and the certificate.

2.3.3 Positioning and extensions

The framework is built on CEB and inherits its residual certificate; the difference is exclusively in the reference, which now carries a declared geometry. With respect to the two failure channels of Section 2.2, the first is settled background: the Caveats paper repairs the dead knob with a squared compression term, and our anchored reference supplies the same strict convexity by a different mechanism — a quadratic displacement term built into the KL by construction, so on the aligned surrogate with the main model's anchor-energy classifier fixing the classifier scale, the training Lagrangian is strictly convex jointly in the alignment and scale variables with a unique optimum per β (Proposition A.1, Appendix; Proposition 2.10). This is a mechanism connection to the Caveats fix, not a new solution of the deterministic degeneration: we do not claim that the β-map is monotone or one-to-one — that would require the full two-dimensional stationarity analysis — and the β-sweep is reported empirically. For the second channel, the accurate positioning is that *OPB removes the unspecified inter-class geometry of the CEB reference*: the reference declares the relationships between labels, so any representation whose class geometry differs from the declared one pays a positive certificate (Propositions 2.9 and 2.3). It does not solve certificate blindness in general — the certificate remains a distance to the reference, not a meter of retained label information. The geometric prior is trainable — the frame rotates and the axis turns — and this rotation channel is *not* among the problems addressed in this paper; we record it as a scope limitation in the appendix. The certificate of Eq. (8) remains the usual CEB residual bound for this particular prior family — we do not claim a tighter mutual information bound — and the isometric axis likewise trades a general conditional prior for a checkable label-metric structure. For multi-dimensional regression labels

$y \in \mathbb{R}^m$ ($m \leq d$), the same construction extends verbatim: take $W \in \mathbb{R}^{d\times m}$, its thin polar factor $Q = W(W^\top W)^{-1/2}$, and the prior $\mu_p(y) = \rho\, Q\, \tilde{y}$, which is isometric in the label metric; the scalar case is $m = 1$ with $Q = u$. The label metric must be declared before standardization, otherwise "isometric" has no unique meaning.

3 Experiments

The experiments address two questions. First, does replacing CEB's unconstrained reference with a constructed geometry — an orthogonal class frame or an isometric axis — cost or gain anything in end-to-end prediction across heterogeneous settings? Second, does prediction survive when the prediction rule is tied to that geometry by construction — the anchor-energy classifier and the tied projection head — so that the free head's degrees of freedom are removed? The main sweep answers both on twelve dataset–backbone settings; the compression–accuracy sweep over β and the anchor scale, and the mechanism-level validation (prior geometry, posterior transfer, nearest-anchor sufficiency, memorization, adversarial robustness), are reported in the supplementary analysis.

3.1 Experimental setup

Datasets. Twelve settings over nine datasets and four backbones, spanning image, graph, and text modalities. *Classification*: MNIST (MLP and CNN backbones); ImageNet-100 (MLP and CNN) on frozen ResNet-50 features (layer-3 activations pooled to 4×4); Cora, transductive node classification (GCN); IMDb sentiment (RNN); AG News (RNN) on frozen BERT layer-10 token features reduced by PCA to 50 dimensions. *Regression*: California Housing (MLP); STS-B (RNN with frozen GloVe-100d embeddings); ZINC-12k molecular graphs (GCN, the official 10k/1k/1k split); AgeDB face age (MLP and CNN). Where no official split exists we use a fixed $60/20/20$ train/validation/test partition (seed 42); on MNIST 10k of the 60k training images are held out for validation. Regression targets are standardized to $[0, 1]$ on the training set (the test metric is re-normalized).

Backbones and protocol. The MLP uses hidden layers 512–256; the CNN two convolutional blocks (channels 32, 64) followed by a 256-dim hidden layer; the GCN matches the MLP sizes; the RNN is a two-layer LSTM with 512 and 256 units (a single 256-unit layer with max time-pooling on STS-B). All methods share the backbone, dropout 0.2, and bottleneck dimension $d = 256$. Every run trains with Adam (learning rate 10^{-3}; 10^{-2} on Cora), at most 100 epochs with early stopping on the validation metric (patience 10), batch size 512 (128 on ZINC), and evaluates the checkpoint with the best validation score on the test set. Every setting is run five times (seeds 0–4); the tables report the mean over the runs. Classification early-stops on validation macro-AUC and reports test accuracy; regression early-stops on validation R^2 and reports test R^2.

Baselines. The plain backbone (Base); VIB [Alemi et al., 2017] with the fixed $\mathcal{N}(0, I)$ prior; SVIB, the squared compression term of Kolchinsky and Tracey [2019] ($\mathrm{CE} + \beta\ \mathrm{KL}^2$); NIB [Kolchinsky et al., 2017], the noise-injection bottleneck with the pairwise-distance upper bound on $I(X; M)$; CEB [Fischer, 2020] with the trainable label-conditional backward encoder; and the supervised β-DVCCA of Abdelaleem et al. [2025], which augments the VIB loss with an input reconstruction term, $\mathrm{CE} + \beta\,(\mathrm{KL} + \mathrm{MSE}_{\mathrm{recon}})$. All variational methods share the encoder, the posterior heads, and the loss convention of Eq. (7).

Our variants. *GPB* denotes our method in the tables. On classification rows it is ***OPB***, the orthogonal-frame prior of Eq. (21) with the anchor-energy classifier of Eq. (27) — the prediction rule has no free parameters, its means and variances are exactly the prior's, and $K \leq d$ holds in all settings ($\max K = 100$ on ImageNet-100); on regression rows it is ***EPB***, the isometric prior of Eq. (28) with the tied projection head $\hat{\tilde{y}} = u^\top z/\rho$. ***OPB-L*** is the free-head ablation: the identical prior and KL, with an ordinary trainable linear classifier (or regression head) on z, so prediction is free to bypass the constructed geometry. The frame is orthonormalized from the all-class mean matrix at every forward pass (sign-normalized reduced QR); the prior encoder is identity-initialized (initial anchors $e_1, \ldots, e_K$, initial KL $\approx a^2/2$ under unit variances), the variance heads are zero-initialized, and the construction is stateless — no EMA, no prototypes, no batch statistics.

Hyperparameter selection. Each method selects its hyperparameters on the validation set from a shared grid: a six-value log-spaced β grid $\{10^{-4}, \ldots, 10\}$ for every variational baseline, and, for OPB and OPB-L, an additional nine-value anchor-scale grid $\{1, \ldots, 16\}$ (6×9 combinations). The main table reports, per method and setting, the test metric of the best validation configuration, so all methods are compared under an identical selection protocol (Base has no sweep and early-stops as usual).

3.2 Main results

Table 1 reports test accuracy (classification) and test R^2 (regression), averaged over five runs, for each method's best configuration; bold marks the best entry per row and underline the second best, and the vertical rule separates GPB and its free-head ablation OPB-L from the baselines. Standard deviations are omitted for width and are available in the release. Three observations.

The constructed geometry is free, and it usually pays. GPB is best or tied-best in 7 of 12 settings with mean rank 1.75, against 4.25 for the plain backbone and 5.0–6.5 for the variational baselines, none of which is ever best. Both GPB and OPB-L dominate CEB — the identical objective with the unconstrained reference — in all 12 settings, and GPB is best among all methods on seven rows. The gains concentrate where the label structure is rich or supervision is scarce: $+3.1$ accuracy over CEB on ImageNet-100 with the CNN backbone (86.1 vs. 83.0), $+1.9$ on Cora, and $+0.08$ R^2 on AgeDB over the plain CNN (0.470 vs. 0.388). On saturated settings (MNIST) the margin is small but consistent, and on the three rows where the plain backbone

Table 1: Test accuracy (%) on classification tasks and test R^2 on regression tasks (mean over 5 runs). Each entry reports the best configuration over the tuned β (and anchor-scale) grid; the best entry per dataset is bolded.

Dataset	Backbone	Base	VIB	SVIB	NIB	CEB	DVCCA	GPB	OPB-L
MNIST	MLP	98.33	98.28	98.15	98.27	98.43	98.24	**98.54**	98.43
MNIST	CNN	99.13	99.28	99.23	99.22	99.30	99.29	**99.32**	99.31
ImageNet-100	MLP	83.12	82.84	83.02	82.35	82.77	82.99	**84.40**	84.01
ImageNet-100	CNN	83.40	82.94	83.11	82.91	82.97	83.21	**86.08**	84.88
Cora	GCN	87.68	86.83	86.46	86.35	86.27	86.90	**88.12**	87.86
IMDb	RNN	84.85	85.50	85.99	86.14	85.55	85.69	86.84	**86.99**
AG News	RNN	**92.94**	92.72	92.86	92.91	92.83	92.78	92.87	92.92
Cal. Housing	MLP	0.7618	0.7901	0.7897	0.7862	0.7894	0.7878	0.7895	**0.7914**
STS-B	RNN	**0.2899**	0.2719	0.2697	0.2417	0.2668	0.2713	0.2762	0.2755
ZINC	GCN	**0.6348**	0.6339	0.6325	0.6178	0.6257	0.6294	0.6344	0.6323
AgeDB	MLP	0.2180	0.3173	0.3385	0.2710	0.3317	0.3180	**0.3536**	0.3320
AgeDB	CNN	0.3914	0.3342	0.3323	0.3337	0.3882	0.3205	**0.4700**	0.4388
Mean rank (of 12)		4.25	5.25	5.00	6.50	5.46	5.50	1.75	2.29
Best or tied-best (of 12)		3	0	0	0	0	0	7	2

remains best (STS-B, ZINC, AG News) the gap is marginal (≤ 0.03) — the geometry is neutral there rather than harmful, and two of the three are regression tasks where every bottleneck method trails the plain backbone.

Prediction does not require the free head. OPB-L, the free linear head on the same geometric prior, is best or tied-best in 2 of 12 settings (mean rank 2.29), and the GPB–OPB-L gap is within 0.4 accuracy or R^2 points in 11 of 12 settings (the exception is ImageNet-100 CNN, 1.2 points): the zero-parameter anchor-energy classifier — whose weights are the frame itself — reaches the accuracy of a freely trained head in almost all settings, so the degrees of freedom removed by construction are not required for prediction, as claimed in the introduction. The three settings where OPB-L edges out GPB (IMDb, AG News, Cal. Housing) confirm that the tie does not inflate the numbers. On the five regression rows the tied projection head of EPB wins or matches the free regression head in four (the exception is Cal. Housing by 0.002).

The tie regularizes the high-cardinality and low-data regimes. The two settings with the largest GPB margins — ImageNet-100 ($K = 100$) and AgeDB (a small, high-variance regression task) — are exactly those where an unconstrained classifier can overfit its own scale or the scarce supervision: fixing the prediction rule to the prior's geometry removes both escape routes, consistent with the fixed-scale condition under which the aligned surrogate of Proposition A.1 has a unique interior optimum per β.

4 Related work

A Deferred proofs and further results

The following results support the main text: the pointwise KL decomposition (Lemma A.1), a Fano-type label-information lower bound (Corollary A.1), the strict convexity of the β-sweep on the aligned family (Proposition A.1), and the scope and limitations (Remark A.1).

Lemma A.1 (OPB KL decomposition)**.** *For a diagonal posterior covariance and the prior of Eq.* (21)*,*

$$\mathrm{KL}\big(q_\phi(z|x)\,\|\,p_\theta^{\mathrm{OPB}}(z|y)\big) = \frac{1}{2\tau^2}\|\mu_\phi(x) - a\,q_{\theta,y}\|^2 + \frac{1}{2}\sum_{r=1}^{d}\Big[\frac{\sigma_{\phi,r}^2(x)}{\tau^2} - 1 - \log\frac{\sigma_{\phi,r}^2(x)}{\tau^2}\Big], \tag{29}$$

a class-anchor mismatch term plus a variance mismatch term that is non-negative and vanishes exactly at $\sigma_{\phi,r}^2(x) = \tau^2$ for all r. The mismatch term is quadratic in the displacement to a reference whose scale and covariance cannot move: this is the mechanism behind the strict convexity of Proposition A.1 below.

Proof. Apply the diagonal-Gaussian KL formula with posterior mean $\mu_\phi(x)$, prior mean $a\,q_{\theta,y}$, posterior variances $\sigma_{\phi,r}^2(x)$, and the fixed prior variance τ^2. □

Corollary A.1 (Label-information lower bound in the aligned Gaussian model)**.** *Assume uniform classes and exact alignment $q(z|Y=k) = \mathcal{N}(a\,q_{\theta,k}, \tau^2 I_d)$. The pairwise error probability of the nearest-anchor classifier is $P_{\mathrm{pair}} = \Phi\big(-\frac{a}{\sqrt{2}\,\tau}\big)$, with $P_e \le \min\{1, (K-1)P_{\mathrm{pair}}\}$, and whenever the displayed bound $\bar{P}_e \le 1 - 1/K$, Fano's inequality gives $I(Y;Z) \ge \log K - h_2(\bar{P}_e) - \bar{P}_e \log(K-1)$. This links the anchor scale-to-noise ratio a/τ to the retained label information, conditionally on the aligned Gaussian model.*

Proof. Two anchors have distance $\sqrt{2}a$ by Proposition 2.8; the equal-covariance binary decision boundary is halfway between them, giving $\Phi(-\sqrt{2}a/(2\tau))$. The union bound and Fano's inequality on $[0, 1-1/K]$ conclude. □

Proposition A.1 (Joint strict convexity and unique optimum on a restricted aligned surrogate)**.** *Consider the class-symmetric aligned family: every input of class y is mapped to the posterior $q_y = \mathcal{N}(s\,q_{\theta,y}, \sigma^2 I)$ ($s \ge 0$, $\sigma^2 > 0$; anchor scale $a > 0$), and the classifier is the anchor-energy classifier of Proposition 2.10 (weights $w_j = c\,q_{\theta,j}$ with scale $c = a/\tau^2$ fixed by construction). With the* stochastic training code $z \sim q_y$ *— as in the actual objective, Eq.* (23) *— the training cross-entropy is*

$$g(s,\sigma^2) := \mathbb{E}_\eta\Big[\ln\Big(1 + \textstyle\sum_{j\neq y} e^{-cs + c\sigma(\eta_j - \eta_y)}\Big)\Big], \qquad \eta \sim \mathcal{N}(0, I_K), \tag{30}$$

and the training Lagrangian is

$$L_\beta(s,\sigma^2) = g(s,\sigma^2) + \beta\Big[\tfrac{1}{2\tau^2}(s-a)^2 + \tfrac{d}{2}\big(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)\big)\Big]. \tag{31}$$

Then for every $\beta > 0$, L_β is strictly convex jointly in (s, σ) and has a unique joint minimizer $(s^*(\beta), \sigma^*(\beta))$, interior in s since $a > 0$: the stochastic training objective has a well-defined optimum, in contrast to the deterministic IB Lagrangian, whose optimum is pinned at a corner for every β (Proposition 2.1). This is a restricted surrogate result*: it analyzes two scalar variables on the class-symmetric aligned family, with the frame, the anchor scale, and the classifier scale held fixed; the full OPB objective — with its nonlinear posterior, trainable frame, and non-convex degrees of freedom — lies outside this analysis, and the result neither establishes β-tunability of the full objective nor makes any claim on the information plane. The deterministic evaluation code $(z = \mu)$ is the $\sigma \to 0$ limit, and $g \geq \ln(1+(K-1)e^{-cs})$ by Jensen (the log-sum-exp is convex), so the statement strictly generalizes the deterministic surrogate analysis. We do* not *claim here that the map $\beta \mapsto s^*(\beta)$ is monotone or that the sweep is one-to-one: the minimizer σ^* responds to β through the cross-derivative $\partial^2 g/\partial s\,\partial\sigma$, and establishing the sign of $\mathrm{d}s^*/\mathrm{d}\beta$ requires the full two-dimensional stationarity analysis, which we leave open; the β-sweep is reported empirically in the experiments. The fixed-scale condition holds for the main model by construction (Proposition 2.10), so the statement describes the actual training objective up to the aligned-family idealization. We do not analyze the free-classifier variant on this surrogate: under the stochastic training code, a diverging classifier scale does not produce a degenerate solution — at $\mathrm{KL} \to 0$ the posterior variance is pinned at $\sigma^2 = \tau^2 > 0$, so the class-conditional Gaussians overlap and the training cross-entropy of a free classifier diverges with its scale rather than vanishing (for $K = 2$ it contains $\mathbb{E}[\ln(1+e^{-c(a+\tau\xi)})]$, whose integrand grows linearly in c on the event $\xi < -a/\tau$, which has positive probability). An analysis of the free classifier would require an explicit classifier-norm constraint or regularizer and is left to future work.*

Proof. For each fixed η, the integrand of Eq. (30) is a log-sum-exp of functions affine in (s, σ) (the constant 1 has slope zero and each exponential has slope $(-c,\ c(\eta_j - \eta_y))$), hence jointly convex in (s, σ), and the expectation g is jointly convex. No strictness is claimed for g: for $K = 2$ the integrand depends on (s, σ) only through the single linear combination $-cs + c\sigma(\eta_2 - \eta_1)$, so its Hessian has rank one. Write the KL term as $R(s, \sigma) := \frac{1}{2\tau^2}(s-a)^2 + \frac{d}{2}\big(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)\big)$; its Hessian in (s, σ) is $\nabla^2 R = \mathrm{diag}\big(\tau^{-2},\ d\,(\tau^{-2} + \sigma^{-2})\big)$, strictly positive definite for every $\sigma > 0$, so βR is strictly convex in (s, σ). Hence $L_\beta = g + \beta R$ is the sum of a jointly convex and a strictly convex function, and is strictly convex in (s, σ). Since $R \to +\infty$ as $s \to \infty$, $\sigma \to 0^+$, or $\sigma \to \infty$, the sublevel sets of L_β are compact, and the unique joint minimizer exists (by strict convexity it is unique). For $a > 0$ the minimizer is interior in s: at $s = 0$, $\partial L_\beta/\partial s = -c\,\mathbb{E}[p_\mathrm{w}] - \beta a/\tau^2 < 0$, where p_w is the error probability of the sampled code, so $s^* > 0$. □

Remark A.1 (The sweep is a KL–CE surrogate curve; scope and limitations). Proposition A.1 proves joint strict convexity and uniqueness of the optimum for the stochastic training objective in the KL–CE plane on the aligned family at fixed classifier scale — a restricted surrogate result, which does not establish β-tunability of the full OPB objective, and in any case not a statement on the information plane of the Caveats paper; the results of the main text establish a structured conditional prior, an exact KL geometry, and conditional separation guarantees, and they do *not* establish a mutual information upper bound tighter than CEB's — Eq. (8) is the usual CEB gap identity for a particular prior family. The orthogonal-frame construction fixes all pairwise anchor distances, so the framework should not claim to recover arbitrary semantic class similarities. All geometric statements are conditional on Assumption 2.1: orthonormality holds exactly by the polar construction wherever the factorization exists, and the condition is verified — not imposed — along the reported runs (Section 2.3.1); the construction is exactly equivariant under relabeling of the classes by Eq. (20), so no class index plays a distinguished role. If the frame were computed from the current input batch rather than from the all-class prior matrix, the prior would become batch-dependent and Proposition 2.4 would no longer follow directly. Finally, the anchor-energy classifier of Proposition 2.10 fixes the classifier scale to a/τ^2 by construction, removing the independent classifier-margin escape route in the aligned surrogate of Proposition A.1; whether a free classifier (with an explicit norm constraint or regularizer) admits its own degenerate route is left to future work, and the free linear classifier is kept in the experiments as a variant for comparison. Whether the joint optimum of Proposition A.1 moves monotonically in β is likewise open: it requires the full two-dimensional stationarity analysis, in which the optimal variance responds to β through the cross-derivative $\partial^2 g/\partial s\,\partial\sigma$ of the stochastic cross-entropy. The trainable frame and axis constitute a rotation channel that this paper does not address: the certificate's zero set contains the full-retention orthogonal label codes at any frame rotation (the reference can rotate to meet them), and the framework pins the *geometry of the code* — it does not claim that the certificate reads off the retained label information. Quantifying how much of the anchor gap is absorbed by frame rotation rather than by posterior motion is left to future work (a frozen-frame ablation isolates it).

References

A. A. Alemi, I. Fischer, J. V. Dillon, and K. Murphy. Deep variational information bottleneck. In *ICLR*, 2017.

I. Fischer. The conditional entropy bottleneck. *Entropy*, 22(9):999, 2020.

A. Kolchinsky and B. D. Tracey. Caveats for information bottleneck in deterministic scenarios. In *ICLR*, 2019.

A. Kolchinsky, B. D. Tracey, and D. H. Wolpert. Nonlinear information bottleneck. *Entropy*, 21(12):1181, 2019.

N. Tishby, F. C. Pereira, and W. Bialek. The information bottleneck method. In *Proc. 37th Allerton Conference*, 2000.

E. Abdelaleem, I. Ndiour, and A. Bhattacharyya. Deep variational multivariate information bottleneck. *JMLR*, 2025.

V. Papyan, X. Y. Han, and D. L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *PNAS*, 117(40):24652–24663, 2020.